

ANSWER KEY — Integration Lesson 36 Test: Arc Length

For the grader · full working shown · give credit for a correct setup with small arithmetic slips · setup-only questions earn full credit with no evaluation

1. Length of $f(x) = (4/3)x$ on $[0, 9]$, checked with the distance formula.

Work: $f' = 4/3 \rightarrow (f')^2 = 16/9 \rightarrow 1 + 16/9 = 9/9 + 16/9 = 25/9 \rightarrow \sqrt{(25/9)} = 5/3$ (a constant, because the slope never changes). $L = \int_0^9 (5/3) dx = (5/3)(9) = 15$. Check: endpoints $(0, 0)$ and $(9, 12)$; distance $= \sqrt{(9^2 + 12^2)} = \sqrt{(81 + 144)} = \sqrt{225} = 15$ ✓ (a 3-4-5 triangle scaled by 3). **Answer: L = 15. Grader note:** for a straight line the integrand is constant, so $L = (\text{constant}) \times (\text{width})$ — no antiderivative work needed.

2. Exact length of $f(x) = 2x - 5$ on $[1, 4]$.

Work: $f' = 2$ (the -5 has derivative 0) $\rightarrow 1 + (f')^2 = 1 + 4 = 5 \rightarrow$ integrand $\sqrt{5}$. $L = \int_1^4 \sqrt{5} dx = \sqrt{5}(4 - 1) = 3\sqrt{5}$. Check: endpoints $(1, -3)$ and $(4, 3)$; distance $= \sqrt{(3^2 + 6^2)} = \sqrt{45} = \sqrt{9 \cdot 5} = 3\sqrt{5}$ ✓. **Answer: L = $3\sqrt{5}$. Grader note:** $3\sqrt{5} \approx 6.71$ is the finished answer — exact form is required, and $\sqrt{45}$ unsimplified should get the point with a note to simplify.

3. Length of $f(x) = (2/3)x^{3/2}$ on $[0, 15]$.

Work: $f' = (2/3)(3/2)x^{1/2} = x^{1/2} = \sqrt{x}$. Then $(f')^2 = x$ (the square undoes the root), so $1 + (f')^2 = 1 + x$ and the integrand is $\sqrt{1 + x}$. Linear inside with slope 1, so plain power rule on the package: $\int \sqrt{1 + x} dx = (2/3)(1 + x)^{3/2}$. $L = [(2/3)(1 + x)^{3/2}]_0^{15} = (2/3)(16^{3/2} - 1^{3/2}) = (2/3)(64 - 1) = (2/3)(63) = 42$. **Answer: L = 42. Check:** $16^{3/2} = (\sqrt{16})^3 = 4^3 = 64$. Sanity: L must be at least the interval width 15 ✓, and at least the chord ≈ 41.5 ✓. **Grader note:** the interval was chosen so $1 + 15 = 16$ is a perfect square — if the student's $16^{3/2}$ is not 64, that's the slip.

4. Length of $f(x) = (2/3)x^{3/2} + 11$ on $[0, 24]$; what does the +11 do?

Work: $f' = \sqrt{x}$ again — the +11 dies when differentiated, and the integrand only ever sees f' . So the integrand is still $\sqrt{1 + x}$. $L = [(2/3)(1 + x)^{3/2}]_0^{24} = (2/3)(25^{3/2} - 1) = (2/3)(125 - 1) = (2/3)(124) = 248/3$. **Answer: L = $248/3$. The +11 lifts the whole curve 11 units up but does not stretch it — a vertical shift never changes arc length (every tiny hypotenuse keeps its size and tilt). Check:** $25^{3/2} = (\sqrt{25})^3 = 5^3 = 125$; $248/3 \approx 82.7 \geq$ width 24 ✓. **Grader note:** a student who tries to keep the 11 inside the integrand has misread the formula; full credit requires both the number and the "shifts don't matter" reason.

5. Set up (do not evaluate) the length of $y = x^2$ on $[1, 3]$.

Work: $f' = 2x \rightarrow (f')^2 = 4x^2$ (square the whole thing, coefficient included) $\rightarrow 1 + 4x^2$. **Answer:** $L = \int_1^3 \sqrt{1 + 4x^2} \, dx$ — stop here. **Grader note:** $(2x)^2 = 4x^2$, not $2x^2$. The parabola has no elementary arc-length antiderivative, so any attempted evaluation is a red flag, not a bonus.

6. Set up (do not evaluate) the length of $y = \sin x$ on $[0, \pi/2]$, and say why we stop.

Work: $f' = \cos x \rightarrow (f')^2 = \cos^2 x \rightarrow$ integrand $\sqrt{1 + \cos^2 x}$. **Answer:** $L = \int_0^{\pi/2} \sqrt{1 + \cos^2 x} \, dx$. We stop because no function built from powers, roots, trig, exponentials, and logs has $\sqrt{1 + \cos^2 x}$ as its derivative — no elementary antiderivative exists, so a calculator must finish it numerically (≈ 1.91). **Grader note:** $\cos^2 x$ means $(\cos x)^2$, not $\cos(x^2)$. Accept any wording of "impossible by hand / elliptic integral / calculator only."

7. Set up (do not evaluate): (a) $y = e^x$ on $[1, 2]$; (b) $y = \ln x$ on $[1, 4]$.

Work: (a) e^x is its own derivative, so $f' = e^x$ and $(f')^2 = (e^x)^2 = e^{2x}$ (multiplying doubles the exponent). (b) $f' = 1/x \rightarrow (f')^2 = 1/x^2$. **Answer:** (a) $L = \int_1^2 \sqrt{1 + e^{2x}} \, dx$ (b) $L = \int_1^4 \sqrt{1 + 1/x^2} \, dx$. **Grader note:** in (b) the form $\sqrt{(x^2 + 1)/x^2}$ — or $(\sqrt{x^2 + 1})/x$, valid since $x > 0$ — is equally correct. Neither integral needs evaluating.

8. $f'(x) = \sqrt{4x^2 + 4x}$: simplify $1 + (f')^2$, then find L on $[0, 3]$.

Work: $(f')^2 = 4x^2 + 4x$ (squaring cancels the root), so $1 + (f')^2 = 4x^2 + 4x + 1 = (2x + 1)^2$ — a perfect square. Re-expand to confirm: $(2x + 1)^2 = 4x^2 + 4x + 1$ ✓. Then $\sqrt{(2x + 1)^2} = 2x + 1$, positive on $[0, 3]$, so no absolute value needed. $L = \int_0^3 (2x + 1) \, dx = [x^2 + x]_0^3 = (9 + 3) - 0 = 12$. **Answer:** $1 + (f')^2 = (2x + 1)^2$, $L = 12$. **Check:** differentiate $x^2 + x \rightarrow 2x + 1$ ✓. Sanity: $12 \geq$ interval width 3 ✓. **Grader note:** arc length only ever needs f' , so being handed the derivative is normal, not missing information.

9. Student's $L = \int_0^{12} \sqrt{(f')^2} \, dx = 5$ for $f(x) = (5/12)x$ on $[0, 12]$. Error? What was computed? Why impossible? Correct length?

Work: (a) The student dropped the 1 inside the square root: $\sqrt{1 + (f')^2}$ became $\sqrt{(f')^2}$. (b) $\sqrt{(f')^2} = |f'|$, so the student integrated the slope and got the total vertical rise of the path: the line climbs $(5/12)(12) = 5$ units. True, but it ignores the 12 units of sideways travel. (c) The integrand $\sqrt{1 + (f')^2}$ is always ≥ 1 , so L must be at least the interval width 12 — an answer of 5 is impossible. (d) Correct: $f' = 5/12 \rightarrow 1 + 25/144 = 144/144 + 25/144 = 169/144 \rightarrow \sqrt{} = 13/12 \rightarrow L = (13/12)(12) = 13$. **Answer:** (a) the 1 was dropped; (b) only the 5 units of rise; (c) $L \geq 12$ always, so 5 is impossible; (d) $L = 13$. **Check:** endpoints $(0, 0)$ and $(12, 5)$: $\sqrt{144 + 25} = \sqrt{169} = 13$ ✓ (the 5-12-13 triangle). **Grader note:** the 1 is the dx^2 leg of the tiny triangle — the horizontal travel. Award the point only if the student explains that, not just fixes the arithmetic.

10. Exact length of $f(x) = x^{3/2}$ on $[0, 5]$, compared with the straight-line endpoint distance.

Work: $f' = (3/2)x^{1/2} \rightarrow (f')^2 = (9/4)x \rightarrow 1 + (f')^2 = 1 + 9x/4$, integrand $\sqrt{1 + 9x/4}$. Linear inside with slope $9/4$, so divide by $9/4$ (i.e. multiply by $4/9$): $\int \sqrt{1 + 9x/4} \, dx = (4/9)(2/3)(1 + 9x/4)^{3/2} = (8/27)(1 + 9x/4)^{3/2}$. (Differentiate to check: $(8/27)(3/2)(9/4)(1 + 9x/4)^{1/2} = (8/27)(27/8)\sqrt{1 + 9x/4} = \sqrt{1 + 9x/4}$ ✓.) At $x = 5$: $1 + 45/4 = 49/4$, and $(49/4)^{3/2} = 7^3/2^3 = 343/8$. At $x = 0$: $1^{3/2} = 1$. $L = (8/27)(343/8 - 1) = (8/27)(343/8 - 8/8) = (8/27)(335/8) = 335/27$. Straight line: endpoints $(0, 0)$ and $(5, 5^{3/2}) = (5, 5\sqrt{5})$; distance $= \sqrt{(25 + 125)} = \sqrt{150} = 5\sqrt{6}$. **Answer: $L = 335/27$, and $335/27 \approx 12.41 > 5\sqrt{6} \approx 12.25$ ✓ — the curve is longer than the chord, exactly as required.** **Grader note:** the interval was designed so $1 + 45/4 = 49/4$ is a perfect square; a student who writes $49^{3/2}/4$ instead of $49^{3/2}/4^{3/2}$ has forgotten to power the denominator too.